

Trigonometric Foundations

SUPPORTING SKILLS LEGEND

Student learning objectives for this unit

I — Introduce

D — Deepen

A — Apply

R — Reinforce

- ☐ 1. I can **use** the Pythagorean theorem to determine an unknown side of a right triangle.

RELATED SKILLS

☐ A Extract perfect-power factors from a radical.☐ R Apply the Pythagorean theorem or its converse.

- ☐ 2. I can **determine** exact side lengths in 30-60-90 and 45-45-90 triangles.

RELATED SKILLS

☐ A Distinguish exact values from measured or rounded approximations.☐ A Extract perfect-power factors from a radical.☐ R Recall special-right-triangle ratios.

- ☐ 3. I can **convert** angle measures between degrees and radians.

RELATED SKILLS

☐ I Convert between degree and radian measure.☐ A Add, subtract, multiply, and divide fractions.

- ☐ 4. I can **determine** coterminal and reference angles for a given angle.

RELATED SKILLS

☐ I Determine a reference angle and its quadrant.☐ I Determine the sign of a trigonometric function by quadrant.☐ I Generate and recognize coterminal angles.

- ☐ 5. I can **calculate** arc length from a central angle and radius.

RELATED SKILLS

☐ D Calculate arc length from radius and central angle.☐ A Convert between degree and radian measure.☐ A Select and report appropriate linear, square, or angular units.

- ☐ 6. I can **determine** exact coordinates on the unit circle.

RELATED SKILLS

☐ I Determine unit-circle coordinates for common angles.☐ A Determine a reference angle and its quadrant.☐ A Recall special-right-triangle ratios.

☐ **7. I can **evaluate** all six trigonometric functions exactly for common angles.**

RELATED SKILLS

- ☐ I Express common trigonometric values exactly.
 - ☐ I Recall the six trigonometric functions, their reciprocal relationships, and their notation.
 - ☐ I Relate sine/cosecant, cosine/secant, and tangent/cotangent.
 - ☐ A Determine the sign of a trigonometric function by quadrant.
 - ☐ A Determine unit-circle coordinates for common angles.
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☐ **8. I can **solve** right triangles using trigonometric ratios.**

RELATED SKILLS

- ☐ D Match opposite, adjacent, and hypotenuse sides to a trigonometric ratio.
 - ☐ D Use a trigonometric ratio to determine a missing side.
 - ☐ D Use an inverse trigonometric function to determine a missing angle.
 - ☐ R Label sides relative to a selected angle.
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☐ **9. I can **apply** right-triangle trigonometry to contextual problems.**

RELATED SKILLS

- ☐ D Match opposite, adjacent, and hypotenuse sides to a trigonometric ratio.
 - ☐ A Attach and interpret units for rates, inputs, and outputs.
 - ☐ A Restrict a model to meaningful contextual inputs.
 - ☐ A Use a trigonometric ratio to determine a missing side.
 - ☐ A Use an inverse trigonometric function to determine a missing angle.
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☐ **10. I can **evaluate** trigonometric functions from a point on a circle or terminal side.**

RELATED SKILLS

- ☐ I Use coordinates and radius to evaluate trigonometric functions.
 - ☐ A Apply the Pythagorean theorem or its converse.
 - ☐ A Determine the sign of a trigonometric function by quadrant.
 - ☐ A Relate sine/cosecant, cosine/secant, and tangent/cotangent.
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☐ **11. I can **solve** a basic trigonometric equation for all angles in a specified interval.**

RELATED SKILLS

- ☐ I Isolate a trigonometric expression before solving.
 - ☐ I Use reference angles and periodicity to find every solution in an interval.
 - ☐ A Determine a reference angle and its quadrant.
 - ☐ A Determine the sign of a trigonometric function by quadrant.
 - ☐ A Translate between inequality and interval notation.
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